

Future trajectories of the BepiColombo MPO and Mio spacecraft: a quantitative tool for magnetospheric region prediction

Daragh M. Hollman, Astronomy & Astrophysics Section, School of Cosmic Physics, Dublin Institute for Advanced Studies, DIAS Dunsink Observatory, Dublin D15 XR2R, Ireland, daragh.hollman@dias.ie

Caitríona M. Jackman, Astronomy & Astrophysics Section, School of Cosmic Physics, Dublin Institute for Advanced Studies, DIAS Dunsink Observatory, Dublin D15 XR2R, Ireland.

Katarina Domijan, Department of Mathematics and Statistics, Maynooth University, Maynooth, Co. Kildare, Ireland.

Charles F. Bowers, Department of Climate and Space Sciences and Engineering, University of Michigan, Ann Arbor, MI, USA.

Lina Z. Hadid, LPP - CNRS, École Polytechnique, Sorbonne Université, Paris-Saclay University, Observatoire de Paris, Paris, France.

Daniel Heyner, Institut für Geophysik und Extraterrestrische Physik, Technische Universität Braunschweig, Braunschweig, Germany.

Abstract

Mercury possesses one of the most dynamic magnetospheres in our solar system, with strong driving by the solar wind, and significant motion of the magnetospheric boundaries in response. The BepiColombo mission will provide the first dual-spacecraft view of this system. In order to maximise the scientific output from BepiColombo, we need quantitative predictions of which magnetospheric region each spacecraft will be sampling at any given time. This is important for the planning of spacecraft operations, where the coordination and collaboration of many instruments across two spacecraft is vitally important, not only for science, but also due to telemetry and power limitations. In this work, we outline a new method using MESSENGER observations to make quantified predictions, with uncertainties, of the probability of observing solar wind, magnetosheath, and magnetosphere for a given position in Mercury’s magnetospheric environment. We apply this technique to five of the six completed BepiColombo Mercury flybys, and additionally forecast magnetospheric region sampling during a representative 10-hour interval of the nominal science mission. We aim to make this approach as accessible as possible, and include in a software repository simple worked examples to aid users in applying these predictions in their own research.

Keywords

Mercury, BepiColombo, MESSENGER, Magnetosphere, Solar Wind, Prediction

1 Introduction

Mercury’s magnetospheric environment has previously been sampled by three missions: Mariner 10, MESSENGER (Mercury Surface, Space Environment, Geochemistry and Ranging), and BepiColombo. Mariner 10 underwent three flybys of Mercury between 1974 and 1975 (Ness et al. 1974). Following three flybys (2008-2009), MESSENGER orbited the planet from 2011-2015 (Leary et al. 2007; Slavin et al. 2008; Baker et al. 2011). BepiColombo will be the first two-spacecraft mission to orbit Mercury, affording unprecedented insights into solar wind-magnetosphere coupling in its dynamic magnetosphere (Milillo et al. 2020; Benkhoff et al. 2021). It is currently on the way to Mercury, having undergone 6 flybys to date. The start of BepiColombo’s orbital insertion is expected in November 2026.

For any spacecraft in a planetary environment, identification of which magnetospheric region (principally solar wind, magnetosheath, or magnetosphere) the spacecraft is located in paramount to contextualise the data observed. This is particularly true for large-scale statistical studies (e.g. Winslow et al. 2013; James et al. 2017; Philpott et al. 2020; Bowers et al. 2024), but also for case study events in which magnetospheric boundary displacements cause regions to deviate from their typical spatial distributions (e.g. disappearing dayside magnetosphere events (Slavin et al. 2019), and Alfvén wing-like configurations (Bowers et al. 2025)). In recognition of the importance of tracking magnetospheric regions (and associated boundary crossings in between), significant effort has gone into manually labelling interactions with the bow shock and magnetopause (Philpott et al. 2020; Sun et al. 2020). Following these works, several authors have employed a plethora of techniques to delve deeper into the thousands of individual boundary crossings during the MESSENGER mission. This included semi-automated criteria (Pump et al. 2025), unsupervised learning (Toy-Edens et al. 2025), and supervised learning (Hollman et al. 2026a). The resulting lists of boundary crossings from such works can be used to determine magnetospheric region for any point in time for the MESSENGER mission. If any given MESSENGER observation occurs after an inbound bow shock crossing, and before an inbound magnetopause crossing, then this observation must be in the magnetosheath. However, this method requires that all of the bow shock and magnetopause crossings have already been identified. Some of these lists of crossings were previously used as the basis for boundary models, specifically static models representing the average shape and position (Winslow et al. 2013; Philpott et al. 2020; Pump et al. 2025). While these models provide an excellent baseline for an empirically-derived average shape and position of these boundaries, the reality of the system is very dynamic and crossings can happen far from these nominal positions. Region prediction based solely on static models generally results in binary classifications (i.e. inside or outside of a given boundary) and cannot statistically quantify the dynamics of Mercury’s magnetosphere.

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Region prediction is important for the coordination of the two-spacecraft BepiColombo mission. MPO (Mercury Planetary Orbiter) and MMO/Mio (Mercury Magnetospheric Orbiter, officially named Mio) will orbit Mercury on very different trajectories to each other (and to MESSENGER), offering the first multi-viewpoint observations of Mercury’s magnetosphere (Benkhoff et al. 2021; Saito et al. 2021). Advance identification of events for scientific cooperation between the two spacecraft is important for

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many reasons, including identifying particularly scientifically interesting spacecraft conjunctions. Examples of such conjunctions include (but are not limited to): (i) times where one spacecraft is near the magnetopause boundary, while the other is deep in the magnetosphere, exploring dayside and nightside reconnection simultaneously, or (ii) times where one spacecraft is in the solar wind, and the other is near the magnetopause boundary, exploring coupling of external solar wind drivers and their effects within the magnetosphere.

In this work, we outline a new method to statistically quantify the likelihood of observing solar wind, magnetosheath, or magnetosphere conditions for any spacecraft in the near-Mercury environment, including appropriate uncertainties on those predictions, based on previous observations from the MESSENGER mission and boundary crossings determined by the Hollman et al. (2026a) region classification model. In Section 2, we describe how a list of previously labelled boundary crossings for the MESSENGER mission can be used to construct spatial maps showing relative region observations - a proxy for the probability of observing each magnetospheric region at a given position. In Section 3, we describe how, by overlaying a given spacecraft’s trajectory on these maps, likelihoods (and uncertainties) for observing each region at each point in time along that trajectory can be determined. We show two examples of this method in practice: an application to five of BepiColombo’s six flybys of Mercury, and an application to MPO and Mio during a representative 10-hour interval of the nominal science mission. Further, in this section we also discuss future improvements, and the caveats and assumptions involved in this method. We summarise this work in Section 4.

2 Methods

In order to make predictions for the BepiColombo mission, or other spacecraft in Mercury’s magnetospheric environment, we first determined which region MESSENGER was located in for the entire mission, and spatially binned these observations to create maps corresponding to the ratio of solar wind, magnetosheath, and magnetosphere in each bin. With these maps, we could then determine the likelihood with quantified uncertainty of observing each region at a given position.

2.1 Determining MESSENGER’s magnetospheric region

In this work, we use a list of crossings by Hollman et al. (2026a), determined automatically using a random forest region classifier. The authors first used this model to automatically predict magnetospheric region during and surrounding boundary crossing intervals by Philpott et al. (2020), before determining crossings between these regions with a series of post-processing steps. This list consists only of an instantaneous timestamp and a label - one of: BS_IN, BS_OUT, MP_IN, or MP_OUT, for boundary type (BS: bow shock, MP: magnetopause), and orientation of the spacecraft trajectory (IN: inbound, OUT: outbound). For example, a BS_IN crossing corresponds to a transition from solar wind to magnetosheath.

Iterating through MESSENGER’s trajectory at a fixed time cadence, we compared each point in time with the list of crossings to determine which region MESSENGER was located in at that time. Ideally, only the nearest crossing in time (before or after) would need to be referenced to determine the current region. If the previous crossing was BS_IN, it follows that MESSENGER was located within the magnetosheath at that time. However, there are some cases which need to be accounted for where crossings in an orbit may be missed. These cases include: data gaps, rare and extreme magnetospheric configurations (e.g. Alfvén wings, Bowers et al. 2025), or potential labelling inaccuracies. Thus, we need to develop a robust system to account for these (rare) anomalies in the dataset. To achieve this, we must instead compare each point along MESSENGER’s trajectory to the nearest crossings both before *and* after, and exclude cases where the crossings disagree. Continuing the previous example, if the following crossing were in fact MP_OUT (instead of the expected MP_IN) the previous crossing would suggest MESSENGER was in the magnetosheath, however, the following crossing would contrast that MESSENGER was in the magnetosphere. These cases are rare ($\sim 0.6\%$ of observations), but need to be accounted for to accurately capture the distributions of region observations.

Using this technique, we iterated through MESSENGER’s trajectory at a cadence of 5 minutes (the choice of which is expanded upon in Section 2.3). We chose not to exclude interplanetary coronal mass ejections (ICMEs, labelled for MESSENGER by Winslow et al. (2015, 2017)) and other extreme behaviour in this process to maintain statistical accuracy to all possible behaviour, however it may be useful to explore both average and extreme behaviour in isolation in future work. Throughout this work we used the aberrated

Mercury-Solar-Magnetospheric (MSM') coordinate system, which is centred at Mercury's dipole (offset $\sim 0.2 R_M$ northward of the planet's geographical centre (Anderson et al. 2011; Johnson et al. 2012)). The x-axis points into the flow of the solar wind, aberrated from sunwards due to Mercury's orbital velocity, which is significant with respect to the solar wind velocity (assumed to be 400 km s^{-1}). Mercury's radial velocity component also has an effect on this aberration, though we chose to ignore it for this calculation as any changes due to Mercury's radial velocity are insignificant with respect to the uncertainties of an assumed constant solar wind speed. The z-axis points northward along Mercury's rotation axis, with the y-axis completing the right-handed orthogonal set. In this work, we follow previous standards and calculate the aberration angle daily to save on computation time (Bowers et al. 2024; Hollman et al. 2026a). The resultant dataset is a table of observations for the entirety of the MESSENGER mission, with columns corresponding to: timestamp, $X_{\text{MSM}'}$, $Y_{\text{MSM}'}$, $Z_{\text{MSM}'}$, $\text{CYL}_{\text{MSM}'} = \sqrt{Y_{\text{MSM}'}^2 + Z_{\text{MSM}'}^2}$, and region label. This dataset was made publicly available in an online data repository (Hollman et al. 2026b).

2.2 Creating maps of relative region occurrence

To explore the spatial distributions of MESSENGER's observations, we constructed 2D heat-maps in the cylindrical plane (X-CYL) under the assumption that Mercury's magnetosphere is cylindrically symmetric (discussed further in Section 3.3). We binned the region observations from Section 2.1 onto a grid of cells with side-length $0.25 R_M$. This cell size was empirically chosen to be small enough to minimise changes between adjacent cells, but large enough to maintain a statistically significant number of observations in each cell. Figure 1 shows these maps. Figure 1a approximates the time spent by MESSENGER in each bin by multiplying the total number of region observations recorded in each bin by the time cadence of observations. Figures 1b-d show the relative region occurrence (RRO, a proxy for region probability) of solar wind, magnetosheath, and magnetosphere respectively, determined by dividing the number of counts of a given region by the total in each bin. For clarity, we shaded cells without observations of a given region grey, but left areas unexplored by MESSENGER as white.

Despite the coarse grid used, the average location of the bow shock and magnetopause can be visually identified at the contour $\text{RRO} = 0.5$. We compared this contour with the average boundary locations

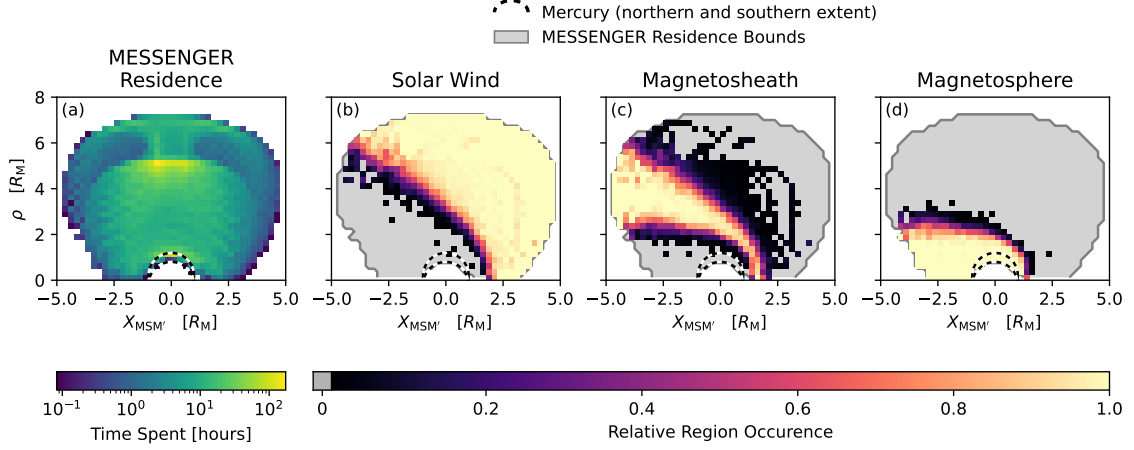


Figure 1. (a) Residence time per spatial bin for the duration of the orbital phase of the MESSENGER mission. (b-d) Relative occurrence of solar wind, magnetosheath, and magnetosphere in each spatial bin. Bins are squares of side length $0.25 R_M$ in the cylindrical plane, $\rho = \sqrt{Y_{\text{MSM}'}^2 + Z_{\text{MSM}'}^2}$, of the aberrated Mercury-Solar-Magnetospheric (MSM') coordinate system. Bins without observations of a given region are shaded in grey - MESSENGER did not observe in white bins. The northern and southern extents of Mercury (as folded into the cylindrical plane) are denoted by white and black dashed lines.

by Winslow et al. (2013) and found average distances of $(0.12 \pm 0.09) R_M$ and $(0.2 \pm 0.1) R_M$ for the bow shock and magnetopause respectively, with best performance at the nose of the boundaries, but positioned too far planetward at the flanks where MESSENGER spent less time. Some artifacts and abnormalities are present in these maps, most notably a singular orbit of magnetosheath extending far into a region of near-always solar wind - appearing in this figure as a vertical line at $X_{\text{MSM}'} \sim 3$ in panels (b) and (c). Mercury's magnetosphere is very dynamic, and the Hollman et al. (2026a) model results show that the magnetopause and bow shock locations can be highly variable in response to extreme solar wind. Furthermore, we know that the most unusual solar wind configurations can result in significant motion, reformation, and even disappearance of the magnetospheric boundaries (e.g. Alfvén wing-like configurations reported by Bowers et al. (2025)). This orbit (and other similar abnormalities) are physically accurate, but represent very extreme cases in spatial regions where the number of observations was small. In this particular example (for bins: $3 R_M \leq X_{\text{MSM}'} < 3.25 R_M$, $2 R_M \leq \sqrt{Y_{\text{MSM}'}^2 + Z_{\text{MSM}'}^2} < 4 R_M$), the time spent in each bin over the entire MESSENGER mission was relatively small (~ 13.5 h on average), and the magnetosheath observations comprised $\sim 3\%$ of

the sample. Additionally, there appears to be some noise at the flanks of the boundaries, where the previously well-defined contours become more diffuse. This was most likely due to a lack of sufficient coverage (as seen in panel (a)), but could also have been due to physical differences in the flanks of Mercury’s magnetosphere.

2.3 Prediction Confidence

To determine a metric of prediction confidence, we treated the prediction of each region (solar wind, magnetosheath or magnetosphere) in each spatial bin as a Bernoulli trial process, with ‘successes’ corresponding to observations of that region, and ‘failures’ corresponding to other region observations. For example, if in a given bin we want to quantify the probability of observing solar wind, we compare the number of observations of solar wind to those of magnetosheath and magnetosphere combined. A confidence interval can then be estimated with a Wilson interval (Wilson 1927), chosen for its performance when sample sizes are small (Brown et al. 2001). A drawback of this approach is that it requires each observation to be statistically independent. We can safely assume this to be true for different orbital passes through the same spatial bin, however, it does not hold true for consecutive measurements on a single orbit. To attempt to satisfy this requirement, we chose to iterate along the trajectory at a time cadence large enough that we can assume observations are independent. We know Mercury’s Dungey cycle operates on timescales of several minutes (Slavin et al. 2021). Additionally, previous studies of the solar wind at Mercury have determined that on timescales longer than 20 minutes, there are likely to be significant differences in the interplanetary magnetic field (James et al. 2017). We therefore chose to iterate through MESSENGER’s trajectory at a cadence of 5 minutes, a balance of minimising the distance between observations to yield greater spatial detail, whilst maintaining that the observations could be considered statistically independent. We note that while these uncertainties are statistically accurate, they are predominantly driven by spacecraft residence and serve to indicate bins which were less sampled by MESSENGER, and hence may be less accurate.

3 Results and Discussion

Taking these maps of relative region occurrence, it is then possible to overlay trajectories on these maps to determine a time series of region predictions for any spacecraft in the Mercury environment. In this

section, we discuss two example applications of this method: one to predict magnetospheric regions for each BepiColombo flyby of Mercury (2021-2025) and another comparing future observations between MPO and Mio during the nominal science mission in 2027. Alongside this work, we provide a software repository with worked examples and accessible tools, to enable any user to easily create similar outputs for trajectories or times of their choosing (Hollman et al. 2026b).

3.1 Example application to BepiColombo flybys of Mercury

We include example probability calculations for 5 of the 6 BepiColombo flybys of Mercury in Figure 2. Flyby 5 was excluded as the trajectory was wholly outside the bounds of MESSENGER’s coverage (closest approach at 37,626 km). Figures 2a-e show time series of the region probabilities determined, spanning two hours before and after closest approach. Times where BepiColombo was outside of the extent of MESSENGER’s spatial coverage were left blank. As the spacecraft approached the edges of MESSENGER’s coverage, the 95% confidence interval becomes wider, reflecting the reduced number of MESSENGER observations in those bins. We note that the apparent size of Mercury’s magnetosphere in this figure (as observed as a plateau of high magnetosphere probability) is not only a function of solar wind conditions, but also of trajectory and velocity, and therefore these panels should not be directly compared. Figures 2f-j show the trajectories for these flybys in the cylindrical plane. We mark closest approach with a circle, and place triangles at fixed time intervals on either side marking direction of travel. These direction markers additionally correspond to vertical lines in panels (a-e) to enable easier comparison between the two. This figure serves as a reference to compare with instruments which collected data during each flyby.

In this example we presented the region probabilities in their raw form, taking the value of the underlying bin for each point in time. This results in a blocky, histogram-like output due to the sharp changes when crossing bin cells - the extent of which is a function of grid cell-size. We could instead smooth this output (or alternatively smooth the underlying probability maps) with simple methods, such as a moving average, to yield a more continuous prediction. This is explored further with examples in the software repository (Hollman et al. 2026b).

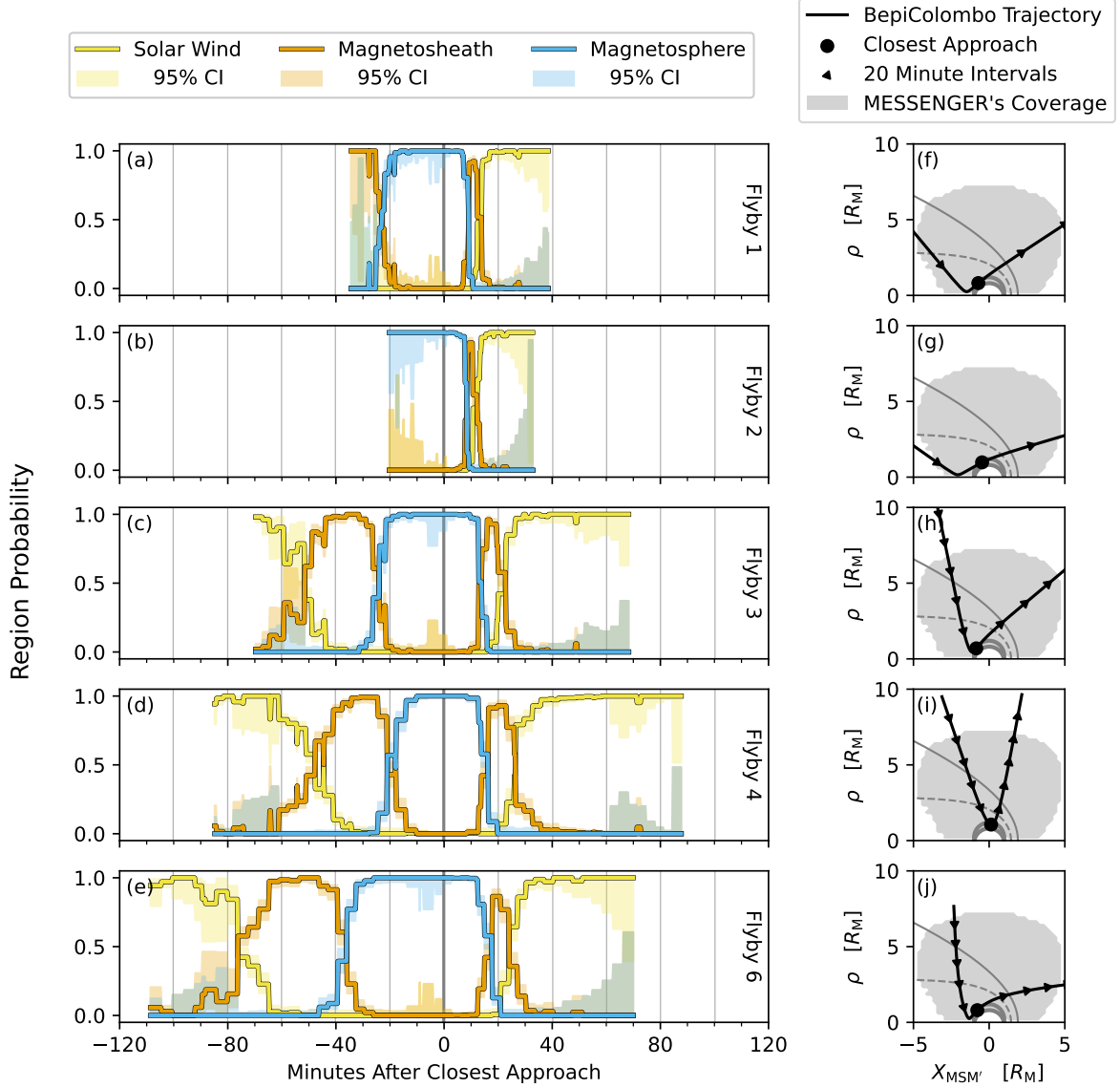


Figure 2. (a-e) Region predictions for 5 flybys of Mercury by BepiColombo (flyby 5 was excluded as it was entirely outside of MESSENGER's spatial coverage). A 95% confidence interval is shaded surrounding each line. These panels are aligned on closest approach (marked by a thick vertical line), with further vertical lines placed on either side at 20 minute intervals. (f-j) Trajectories of these 5 flybys as projected to the cylindrical plane, $\rho = \sqrt{Y_{MSM'}^2 + Z_{MSM'}^2}$. For each flyby, BepiColombo's closest approach is marked by a black circle, with triangles placed at 20 minute intervals either side - additionally marking the travel direction. The northern and southern extent of Mercury (as projected to this plane) are drawn in dark grey, along with average bow shock and magnetopause positions (Winslow et al. 2013). The extent of MESSENGER's spatial coverage is shaded in the background.

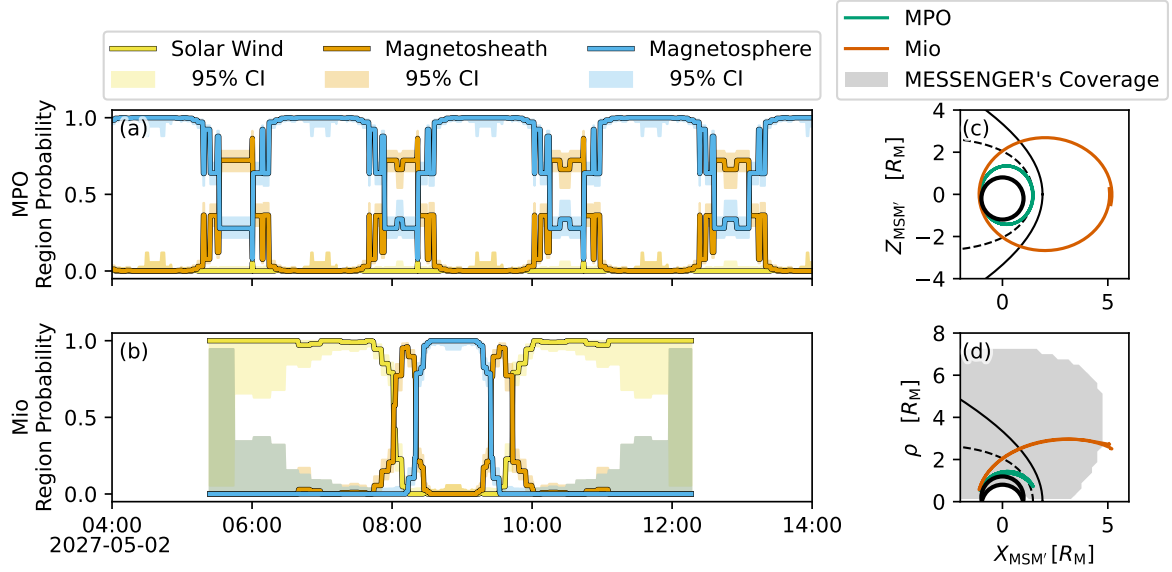


Figure 3. (a, b) Region predictions for both MPO and Mio on 2nd May 2027 from 04:00 to 14:00. Solar wind, magnetosheath, and magnetosphere probabilities are drawn in yellow, orange, and blue respectively, with a 95% confidence interval shaded surrounding them. (c-d) The expected trajectories in the XZ and cylindrical ($\rho = \sqrt{Y_{MSM'}^2 + Z_{MSM'}^2}$) planes of MPO (green) and Mio (red) during this time period - generated with BepiColombo planning SPICE kernel BC_PLAN_V432-20260102-001.TM. MPO orbits Mercury several times during this time period, while Mio undergoes ~ 1 orbit (starting and finishing at the right of panel (d)). MESSENGER's spatial coverage in the cylindrical plane is shaded in grey.

3.2 MPO and Mio region predictions during the nominal science mission

Figure 3 shows example region probabilities for MPO (a) and Mio (b) between 04:00 and 14:00 on the 2nd May 2027. We chose this date to be during the nominal science mission of BepiColombo, and specifically, a time when Mio is predicted to observe all three regions in one orbit. Panel (c) shows the trajectories of the two spacecraft at this time in the cylindrical frame. MPO completes several orbits during this interval, while Mio completes approximately one orbit. As Mio's trajectory approaches the edge of MESSENGER's spatial coverage, the 95% confidence intervals dramatically increase in magnitude. As in Section 3.1, this was due to the lack of MESSENGER observations in those bins.

This figure highlights the utility of this work as a predictive tool. We can determine region probabilities

for MPO and Mio (or any spacecraft in Mercury’s magnetospheric environment) for any time span. If, for example, we were scientifically or operationally interested in finding all times when Mio was in the solar wind and MPO was in the magnetosphere, we could determine these region probabilities for a large span of time and quantify the likelihood of that configuration as a function of time by multiplying $P(\text{MPO in Magnetosphere})$ and $P(\text{Mio in Solar Wind})$, propagating the uncertainties.

3.3 Potential improvements and caveats

The region maps we discussed in Section 2.2 and showed in Figure 1 could be further improved by filtering the set of observations used in their construction. For example, as we know the subsolar point of Mercury’s magnetosphere varies with heliocentric distance (Zhong et al. 2015a; He et al. 2022), we could construct specialised versions of these region maps for use with specific conditions. If, for instance, we were interested in making predictions at a time near perihelion, we could reconstruct the region maps but only include observations within similar heliocentric distances. Similarly, we could use IMF orientation as another filter (in isolation or in tandem with another filtering parameter). In principle, any parameter which can be determined for all time during the MESSENGER mission could be used to filter the region observations to more specific conditions. This observation filtering is a simple yet powerful technique to improve the performance of this method, particularly when used for comparison with existing data, however, its capabilities are obviously limited in a predictive capacity to those parameters which can also be predicted. Additionally, we must be careful to not filter the MESSENGER observations too extensively, as statistical accuracy in the region maps depends on the number of observations in each bin, and if filtered too much uncertainties can grow large. We have ensured that this method is flexible to additional observations, and future observations from both BepiColombo spacecraft will have the potential to expand upon the coverage in these maps, and improve their accuracy within MESSENGER’s coverage. BepiColombo spacecraft will spend $\sim 92.4\%$ ($\sim 99.6\%$ for MPO and $\sim 85.2\%$ for Mio) of time within MESSENGER’s coverage. Outputs of this method are limited to times where spacecraft are within the bounds of MESSENGER’s coverage, though future work could also include expanding these bounds using techniques such as non-linear support vector machines (e.g. Nie et al. 2020) or other methods.

We limited the scope of this work to determine solar wind, magnetosheath, and magnetosphere regions.

These three categories cover a broad range of more detailed ‘sub-regions’, such as the foreshock region (downstream of the quasi-parallel bow shock) or regions within the magnetosphere such as the plasma-sphere. Grison et al. (2025) manually labelled 13 individual region types for each spacecraft of the Cluster mission at Earth. Plasma instruments onboard enabled a more detailed view of the magnetospheric environment, allowing more detailed labelling. While MESSENGER’s plasma instruments were somewhat limited in the context of region determination (see Toy-Edens et al. 2025; Hollman et al. 2026a, for more details), this work could be revisited during BepiColombo’s nominal science mission with more detailed labelling of MPO and Mio observations to produce a prediction tool for more than these three broad region types.

In the development of this work, we made several assumptions which are important to discuss. Perhaps most notably, throughout this work we have folded all observations and trajectories into the cylindrical plane under the assumption that Mercury’s magnetospheric boundaries are cylindrically symmetric (as similarly done in: Winslow et al. 2013; Philpott et al. 2020; He et al. 2022, and others). This was an important assumption as it vastly improved MESSENGER’s spatial coverage around Mercury. If we were instead to use a 3D grid, the density of observations would have been much lower, and consequentially we would have required a larger grid cell-size to achieve the same level of accuracy in region probability (reducing the spatial accuracy). While we believe assumption is reasonable in this application, it is important to note its physical inaccuracies. Zhong et al. (2015b) showed that Mercury’s magnetopause has north-south asymmetries due to the northward offset in Mercury’s dipole. They additionally showed that Mercury’s magnetopause is elongated in the north-south direction tail-ward of the terminator, likely resulting in increased uncertainties for predictions made near the flanks.

Other caveats to consider include that BepiColombo’s orbits will be separated from MESSENGER’s by roughly 90° in the Z_{MSM} axis for the same true anomaly angles, potentially yielding different conditions for the same spatial regions sampled. We may also expect to see differences due to solar cycle. MESSENGER orbited Mercury during the maximum of the 24th solar cycle, while BepiColombo’s nominal science mission will occur during the declining phase of the 25th. The average conditions and frequency of extreme events will potentially differ from what MESSENGER observed. These effects may result in a less reliable output when making predictions for BepiColombo. The extent of these cannot

be quantified at this time, however, following the arrival of the BepiColombo mission at Mercury we can utilise the new observations from MPO and Mio to better understand these effects. A roadmap of future work to explore this may include: (a) First identifying bow shock and magnetopause crossings in MPO and Mio data, either manually, or through the application of automated methods (e.g. Hollman et al. 2026a, or similar) - though the application of such methods to BepiColombo are currently untested. (b) Statistically comparing region observations from MPO and Mio to the predictions determined by this work and quantifying any differences. (c) Incorporating region observations from MPO and Mio into the probability maps shown in Figure 1 to expand the prediction bounds and reduce uncertainties.

4 Summary

In this work we have outlined a new method to make quantified predictions of solar wind, magnetosheath, and magnetosphere for any spacecraft in Mercury’s magnetospheric environment. This work utilised previous labelling of MESSENGER bow shock and magnetopause crossings to determine a dataset of observations of magnetospheric region for the entirety of the MESSENGER mission. We binned those observations spatially in the cylindrical plane of the Mercury-Solar-Magnetospheric coordinate system to serve as maps of relative region occurrence with quantified uncertainties. By drawing trajectories of BepiColombo spacecraft over these maps and picking out the map values, we showed how time series estimating region probability for a given span of time could be determined. We demonstrated two examples of this work in use: across five of BepiColombo’s six flybys of Mercury, and additionally an example during the nominal science mission (i.e. post-separation of MPO and Mio). These two examples exist in the software repository (Hollman et al. 2026b), along with other simple worked examples to aid users in creating their own predictions. We additionally outlined the caveats and assumptions associated with this method, and discussed how it could be further improved in future work.

Availability of data and materials

The datasets generated and/or analysed during the current work are available in the following Zenodo repositories:

- daraghollman/bepi-region-prediction: <https://doi.org/10.5281/zenodo.18348157> (Hollman et al. 2026b)

- MESSENGER Magnetospheric Boundary Crossings: <https://doi.org/10.5281/zenodo.15706295>

BepiColombo SPICE kernels are publicly available at the ESA SPICE Service HTTP:

- <http://spiftp.esac.esa.int/data/SPICE/BEPICOLOMBO/kernels>

Competing interests

The authors declare no competing interests.

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Authors' contributions

Author contributions are as follows:

Conceptualization: DMH, CMJ, KD, CFB.

Data curation: DMH.

Methodology: DMH, CMJ, KD, CFB.

Software: DMH.

Visualisation: DMH, CMJ, CFB.

Writing - original draft: DMH.

Writing - review & editing: DMH, CMJ, KD, CFB, LZH, DH.

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